

Nocardiosis: A Comprehensive Disease Characteristics Report

Disease: Nocardiosis · **MONDO:** MONDO:0017776 · **Category:** Infectious Disease **Report type:** Disease knowledge-base template (15 sections) · **Iterations completed:** 5 · **Findings confirmed:** 8 · **Papers reviewed:** 59 **Evidence base:** Literature synthesis (PubMed). No patient-level data files were provided; all content is derived from aggregated disease-level literature (systematic reviews, multicenter cohorts, case series), not individual EHR data.

Summary

Nocardiosis is an **opportunistic bacterial infection** caused by aerobic, Gram-positive, weakly (partially) acid-fast, branching filamentous actinomycetes of the genus *Nocardia*. It is an **environmentally acquired, non-heritable infectious disease**: the organism lives in soil, decaying vegetation and water, and enters the host by **inhalation** (producing pulmonary disease, the primary presentation) or by **traumatic inoculation** (producing cutaneous/subcutaneous disease, including actinomycetoma). Its defining clinical feature is a strong propensity for **hematogenous dissemination to the central nervous system**, where it forms ring-enhancing brain abscesses carrying a case-fatality of roughly 20–30%. Because the etiology is infectious rather than genetic, the many template sections that assume a Mendelian/heritable disease (causal human genes, inheritance pattern, penetrance, germline variant classification, genetic screening) are **Not Applicable**; the molecular characterization instead centers on **pathogen virulence factors** and **antimicrobial-resistance determinants**.

The disease disproportionately afflicts hosts with impaired **cell-mediated immunity**. The dominant risk factors — corticosteroid therapy, solid-organ and hematopoietic-cell transplantation, high calcineurin-inhibitor (tacrolimus) exposure, low CD4 counts (HIV/AIDS), and anti-GM-CSF autoantibodies — all converge on defective T-cell/macrophage killing. In solid-organ transplant recipients nocardiosis affects 0.04–3.5% of patients and confers a roughly 10-fold higher one-year mortality than in transplant recipients without infection. Diagnosis rests on demonstrating branching, beaded, weakly acid-fast filaments on microscopy, growth on culture (slow, days to weeks), and species-level identification by MALDI-TOF MS, 16S

rRNA gene sequencing, MLSA, or whole-genome/metagenomic sequencing. Chest CT typically shows nodules/masses, consolidation and cavitation; brain MRI shows ring-enhancing abscesses with vasogenic edema.

Treatment is prolonged (months) and **species- and susceptibility-guided**. **Linezolid is universally active** against *Nocardia*; trimethoprim-sulfamethoxazole (TMP-SMX) and amikacin cover >90% of isolates and form the backbone of therapy, with carbapenems and third-generation cephalosporins used for severe/disseminated disease and neurosurgical drainage for brain abscesses. Resistance is species-specific and gene-encoded (e.g., *bla*FAR-1 in *N. farcinica*, *bla*AST-1 in *N. cyriacigeorgica*, *gyrA* mutations, 16S rRNA methyltransferases). Prevention is chiefly **dose-dependent TMP-SMX chemoprophylaxis** in high-risk immunosuppressed patients plus minimization of net immunosuppression; no vaccine exists for humans.

Section 1 — Disease Information

Overview. Nocardiosis is a localized or disseminated opportunistic infection caused by *Nocardia* spp., environmental actinomycetes. It most often begins as a subacute-to-chronic pulmonary infection and can spread hematogenously, with the CNS the most common secondary site, followed by skin/soft tissue. It is not contagious person-to-person.

Key identifiers. - **MONDO:** MONDO:0017776 (nocardiosis) - **ICD-10:** A43 (A43.0 pulmonary, A43.1 cutaneous, A43.8 other forms, A43.9 unspecified); **ICD-11:** 1C1F - **MeSH:** D009617 (Nocardia Infections) - **OMIM / Orphanet:** Not a Mendelian disorder — no OMIM disease number. Orphanet lists nocardiosis as a rare infectious disease. - **Causative genus (NCBI Taxonomy):** *Nocardia* (taxid 1817); key species include *N. asteroides*, *N. farcinica*, *N. cyriacigeorgica*, *N. brasiliensis*, *N. otitidiscaviarum*, *N. nova*, *N. seriolae* (fish).

Synonyms / alternative names. Nocardia infection; nocardial infection; actinomycetoma / Nocardial mycetoma (for the chronic subcutaneous form). Historically many cases were attributed to "*N. asteroides* complex" before molecular taxonomy split it into distinct species.

Information source. The evidence base is a mix of **individual patient reports/case series** (EHR-derived clinical cohorts) and **aggregated disease-level resources** (systematic reviews, multicenter case-control studies). No population-level germline registry applies.

Section 2 — Etiology

Primary cause — infectious. Nocardiosis is caused by infection with *Nocardia* spp. As one review states, "*Nocardia are uncommon pathogens that disproportionately afflict the immunocompromised host*" (PMID: 29668123). Acquisition is by inhalation of soil dust (pulmonary/disseminated disease) or by percutaneous trauma with soil/plant contamination (cutaneous disease, actinomycetoma).

Risk factors (host, non-genetic). Impaired **cell-mediated immunity** is the central predisposing state: - **Corticosteroid therapy** — the single most common predisposing factor; in a systematic review of CNS nocardiosis, corticosteroid use was present in 55.8% of patients (PMID: 35700710). - **Solid-organ transplantation (SOT)** — affects 0.04–3.5% of recipients; independent risk factors include high calcineurin-inhibitor trough levels (OR 6.11, 95% CI 2.58–14.51), tacrolimus use (OR 2.65, 95% CI 1.17–6.00) and corticosteroid dose (OR 1.12 per unit, 95% CI 1.03–1.22) (PMID: 27090987). - **Hematopoietic-cell transplantation (HCT)** and delayed CD4 T-cell recovery. - **High-dose methylprednisolone and CMV infection** (independent risk factors in renal transplant recipients; PMID: 39254072). - **HIV/AIDS with low CD4 counts.** - **Anti-GM-CSF autoantibodies / autoimmune pulmonary alveolar proteinosis** (French multicenter study,).

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- **Chronic lung disease** (bronchiectasis, COPD), **diabetes mellitus**, connective tissue disease, malignancy. - **Sulfonamide allergy label** — associated with higher nocardiosis risk in SOT (HR 3.85, 95% CI 1.44–10.30), likely via avoidance of protective TMP-SMX prophylaxis (PMID: 39136148).

Genetic risk factors (human host). No established Mendelian susceptibility gene. Rare monogenic immunodeficiencies affecting phagocyte/T-cell function (e.g., chronic granulomatous disease) can predispose to *Nocardia*, but there is no disease-causing germline locus. Anti-GM-CSF autoantibody-driven susceptibility is acquired, not inherited. **This subsection is largely Not Applicable.**

Protective factors. **TMP-SMX chemoprophylaxis** reduces risk in a dose-dependent manner (see Section 13). No genetic protective variant is defined.

Gene–environment interactions. Not applicable in the human host in a Mendelian sense. The relevant interaction is **iatrogenic immunosuppression × environmental exposure** (soil/dust) — pharmacologic suppression of cell-mediated immunity converts an environmental saprophyte into a lethal pathogen.

Section 3 — Phenotypes

Phenotypes are **infection manifestations**, not heritable traits. HPO terms below are suggested for knowledge-base mapping.

Phenotype	Type	Frequency / notes	Suggested HPO
Pulmonary infection (pneumonia, nodules, cavitation)	Clinical/ imaging	~64% of cases (28/44) (PMID: 41319540)	HP:0002090 (Pneumonia); HP:0100750 (Pulmonary nodules)
Fever	Symptom	Very common (100% in bacteremia series)	HP:0001945 (Fever)
Cough, dyspnea	Symptom	Common in pulmonary disease	HP:0012735 (Cough); HP:0002094 (Dyspnea)
Brain abscess / CNS involvement	Clinical	Most common secondary site; 86.9% of CNS cases show brain abscess (PMID: 35700710)	HP:0025186 (Brain abscess); HP:0002383 (Encephalitis)
Focal neurological deficit / altered mental status	Sign	In CNS disease	HP:0034332 (Cognitive decline); HP:0002011 (CNS abnormality)
Cutaneous/subcutaneous abscess, actinomycetoma	Physical	Primary cutaneous (immunocompetent) or metastatic (disseminated)	HP:0025503 (Skin abscess)
Disseminated multi-organ infection	Clinical	~30% (13/44) (PMID: 41319540)	HP:0032262 (Disseminated infection)
Weight loss, malaise	Constitutional	Chronic presentations	HP:0001824 (Weight loss)

- **Age of onset:** Predominantly adults; mean age $\sim 55 \pm 16$ y in CNS series ([PMID: 35700710](#)). Not congenital.

- **Severity:** Variable — from indolent cutaneous lesions to fatal disseminated/CNS disease.
- **Progression:** Subacute to chronic; can be rapidly progressive in severely immunosuppressed hosts.
- **Quality-of-life impact:** Substantial in disseminated/CNS disease (neurological sequelae, prolonged multi-month therapy, hospitalization); disease-specific QoL instruments are not established.

Section 4 — Genetic / Molecular Information

Human causal genes: Not Applicable. Nocardiosis is infectious; there is no human causal gene, pathogenic germline variant, ACMG/AMP variant classification, modifier gene, inheritance-linked epigenetic lesion, or chromosomal abnormality. Sections requiring gnomAD allele frequency, somatic vs germline origin, and human variant nomenclature do not apply.

Molecular characterization instead centers on the pathogen genome. *Nocardia* virulence and resistance are gene-encoded (detailed in Sections 5, 6, 12). Genomic analysis of *N. seriolae* and clinical isolates identified virulence-associated genes including *sodA* (superoxide dismutase), *katG* (catalase), and mycolic-acid/cell-wall biosynthesis genes (*pks13*, *fadD32*, *pcaA*, *mftF*) (PMID: 40827537). Resistance determinants include *blaFAR-1*, *blaAST-1*, *aph(2'')*, *gyrA* Ser83Ala, and a 16S rRNA m1A1408 methyltransferase (PMID: 37527397).

Section 5 — Environmental Information

Environmental reservoir. *Nocardia* is a ubiquitous soil saprophyte found in soil, decaying organic/plant matter, dust and water. Exposure to soil and dust (gardening, agriculture, construction) is the principal environmental risk; several case reports link disease to occupational/recreational soil exposure.

Lifestyle factors. Smoking and chronic lung disease increase pulmonary susceptibility; alcohol use and diabetes are frequent comorbidities. These are host-modifying rather than direct causal factors.

Infectious agents (the cause). Human disease is caused by multiple *Nocardia* species. Predominant clinical isolates include *N. asteroides*, *N. farcinica*, and *N. cyriacigeorgica* (PMID: 41319540); *N. brasiliensis* is the classic cause of actinomycetoma; *N. otitidiscaviarum* shows unpredictable susceptibility; *N. seriolae* causes fish nocardiosis. *N. farcinica* is characteristically multidrug-resistant and over-represented in brain/disseminated disease.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Environmental exposure** — the host inhales soil/dust aerosols containing *Nocardia*, **or** the organism is inoculated through broken skin (trauma). → leads to
2. **Deposition in alveoli (or dermis)** and encounter with resident macrophages. → leads to
3. **Phagocytosis by macrophages**, which would normally kill the organism. → but
4. **Immune-evasion branch:** *Nocardia* resists intracellular killing. Secreted **catalase (*katG*) and superoxide dismutase (*sodA*)** detoxify reactive oxygen species; the **mycolic-acid-rich cell wall** (built by *pks13*, *fadD32*, *pcaA*, *mftF*) resists phagolysosomal degradation; secreted **proteases** damage tissue and **immunosuppressive metabolites (e.g., brasiliardin A)** blunt local immunity (PMID: 40827537; PMID: 33956121). → results in
5. **Intracellular survival and replication within macrophages** — favored when host **cell-mediated immunity is impaired** (corticosteroids, calcineurin inhibitors, low CD4). → leads to
6. **Local suppurative/granulomatous inflammation.** In the lung: pneumonia, nodules, consolidation and cavitation. A local **anti-inflammatory (immunosuppressive) microenvironment coexists with a systemic acquired immune response** (PMID: 22825801). Nitric oxide (eNOS) is required for actinomycetoma lesion formation in mice (PMID: 33956121). → then branches 7a. **Containment branch (competent immunity):** granuloma / macrophage barrier walls off the organism → localized, indolent disease. 7b. **Dissemination branch (impaired immunity):** organisms breach the local barrier and enter the bloodstream. → leads to
7. **Hematogenous spread** to the CNS (most common secondary site), skin/soft tissue, and other organs. → results in
8. **Ring-enhancing brain abscess** with surrounding vasogenic edema. → leads to

9. Clinical manifestation: focal neurological deficit, altered mental status, seizures; untreated or delayed, **death** (CNS case-fatality ~23%).

(Steps 4–5 are supported by pathogen genomics and animal models; the human host-side detail is inferred from clinical association with cell-mediated immune defects rather than mechanistically demonstrated in humans.)

Detail by category

- **Molecular pathways / cellular processes:** Macrophage phagocytosis and oxidative burst; ROS detoxification; granuloma formation; chronic inflammation. Suggested GO terms: GO:0006909 (phagocytosis), GO:0000302 (response to reactive oxygen species), GO:0006954 (inflammatory response), GO:0006801 (superoxide metabolic process).
- **Protein dysfunction (pathogen):** Mycolic-acid synthesis enzymes and antioxidant enzymes are gain-of-function virulence assets, not host defects.
- **Immune involvement:** Host defect is impaired **cell-mediated immunity** (T-cell/macrophage axis); IFN- γ -driven macrophage activation is protective (demonstrated in fish models, [PMID: 33045332](#)).
- **Tissue damage mechanisms:** Suppuration/necrosis (abscess), protease-mediated tissue destruction, granulomatous fibrosis.
- **Cell types (CL terms):** macrophage (CL:0000235), neutrophil (CL:0000775), T cell (CL:0000084), CD4⁺ T cell (CL:0000624).

Section 7 — Anatomical Structures Affected

- **Primary organ: Lung** (UBERON:0002048) — the portal of entry for inhaled disease; pulmonary infection in ~64% of cases.
- **Most common secondary organ: Brain** (UBERON:0000955) — brain abscess in the vast majority of CNS cases.
- **Other sites:** skin/subcutaneous tissue (UBERON:0002097), bone/joint/bursa, pericardium, lymph nodes, bloodstream (bacteremia), eye, kidney.
- **Body systems:** respiratory, nervous (CNS), integumentary, and (when disseminated) systemic/cardiovascular.

- **Tissue level:** lung parenchyma (alveolar epithelium and interstitium), brain parenchyma, dermis/subcutis.
 - **Cell populations (CL):** macrophage (CL:0000235), neutrophil (CL:0000775).
 - **Subcellular (GO cellular component):** phagosome/phagolysosome (GO:0045335 / GO:0032010) is the key intracellular niche.
 - **Localization/lateralization:** Pulmonary lesions often multifocal/bilateral; brain abscesses may be single or multiple, unilateral or bilateral.
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Section 8 — Temporal Development

- **Onset:** Predominantly **adult**; subacute to chronic (insidious) course. In SOT, median onset is **17.5 months post-transplant** ([PMID: 27090987](#)). Acute fulminant presentations occur in severe immunosuppression.
 - **Progression:** From localized pulmonary focus to disseminated/CNS disease over weeks to months if untreated. Cavitation is more frequent in immunosuppressed and disseminated disease.
 - **Course pattern:** Progressive without treatment; treatment-responsive but requires **prolonged (weeks to ≥ 6 –12 months) therapy** to prevent relapse.
 - **Duration:** Not self-limited; chronic if untreated.
 - **Critical period:** Early diagnosis and drainage are decisive — delay increases CNS/dissemination and mortality. Prophylaxis interruption during immunosuppression transitions is a recognized trigger.
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Section 9 — Inheritance and Population

- **Inheritance: Not Applicable** — infectious, non-heritable. No inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency.
- **Epidemiology:** Estimated incidence historically ~500–1,000 US cases/year (increasing with immunosuppressed populations); precise population incidence is uncertain and under-reported. In SOT, cumulative incidence 0.04–3.5% ([PMID: 27090987](#)).

- **Demographics: Male predominance** (~70.8% male in CNS series); mean age ~55 y ([PMID: 35700710](#)). Notably, ~34% of CNS cases were immunocompetent, so immunocompetence does not exclude the diagnosis.
- **Geographic distribution:** Worldwide; actinomycetoma (*N. brasiliensis*) is more common in tropical/subtropical regions. Species distribution varies by region (e.g., *N. cyriacigeorgica* common in India, [PMID: 40112638](#)).

Section 10 — Diagnostics

Microbiology (gold standard). Direct microscopy of Gram-stained smears shows **branching, beaded, filamentous Gram-positive rods** that are **weakly/partially acid-fast** on modified Kinyoun stain. Gram-stained smears are emphasized as a valuable, easily-overlooked diagnostic step ([PMID: 36636850](#)). Culture is slow (days to weeks). Species identification uses **MALDI-TOF MS, 16S rRNA gene sequencing, MLSA, WGS, and metagenomic next-generation sequencing (mNGS)** ([PMID: 42258415](#)).

Imaging. - **Chest CT:** Multiple/solitary nodules or masses are the most common finding (94.12% in one series), with ground-glass opacities (76.47%), patchy consolidation (73.53%), and cavitation (52.94%) ([PMID: 37185005](#)). Cavitation is significantly more frequent in immunosuppressed than immunocompetent patients (85% vs 29%, $p = 0.005$, [PMID: 37185005](#)) and in disseminated vs localized disease (64.3% vs 32.8%, [PMID: 38529577](#)). - **Brain MRI/CT (contrast):** ring-enhancing abscess(es) with vasogenic edema.

Susceptibility testing. Broth microdilution (CLSI) for TMP-SMX, linezolid, amikacin, carbapenems, third-generation cephalosporins, amoxicillin-clavulanate, moxifloxacin, minocycline. Species identification predicts likely resistance patterns.

Clinical criteria / differential diagnosis. No formal scoring system; diagnosis is microbiological. Differential includes tuberculosis, actinomycosis, fungal infection (aspergillosis, cryptococcosis), metastatic malignancy and lung cancer — pulmonary and CNS nocardiosis frequently **mimic malignancy** ([PMID: 42007821](#)).

Screening / genetic testing: Not applicable (no germline component).

Section 11 — Outcome / Prognosis

- **CNS nocardiosis case-fatality:** 22.8% overall ([PMID: 35700710](#)); nocardial brain abscesses carry a 20–55% mortality ([PMID: 39780099](#)).
 - **SOT nocardiosis:** one-year all-cause mortality **16.2% vs 1.3%** in matched controls — a **10-fold increase** ([PMID: 28329348](#)); 12-month mortality ~16.8% in a multicenter SOT cohort, with liver transplantation and shorter symptom-to-presentation time independently associated with death ([PMID: 36303280](#)).
 - **Prognostic factors:** immunosuppression intensity, CNS/disseminated involvement, species (*N. farcinica* worse), delayed diagnosis. **Surgery improves survival** in CNS disease (multivariate OR 2.4, 95% CI 0.99–4.11, $p = 0.046$, [PMID: 35700710](#)).
 - **Complications:** brain abscess rupture, respiratory failure, multi-organ dissemination, relapse if therapy is too short.
 - **Recovery:** Good with early, susceptibility-guided, adequately prolonged therapy plus drainage; poor when diagnosis is delayed.
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Section 12 — Treatment

Backbone pharmacotherapy. *Nocardia* susceptibility is favorable to a defined set of agents:

Agent	Activity across isolates	Notes	Suggested NCIT
Linezolid	100% (universally active)	Oxazolidinone; key for CNS/severe/resistant disease	NCIT:C1649
TMP-SMX (cotrimoxazole)	93%	Sulfonamide backbone; oral; prophylaxis + treatment	NCIT:C312
Amikacin	91%	Aminoglycoside; combination for severe disease	NCIT:C233
Carbapenems (imipenem/meropenem)	High	Severe/disseminated, CNS combination	NCIT:C61796
Third-gen cephalosporins (ceftriaxone/cefotaxime)	Species-dependent	<i>N. farcinica</i> often resistant (<i>blaFAR-1</i>)	NCIT:C548

Data: Israeli WGS study of 138 strains — "*Linezolid was active against all isolates, followed by trimethoprim/sulfamethoxazole (93%) and amikacin (91%)*" ([PMID: 37527397](#)); Indian series — "*All tested isolates were susceptible to linezolid and 96% susceptible to amikacin*" ([PMID: 40112638](#)).

Species-specific resistance (gene-encoded): *blaFAR-1* → ceftriaxone resistance in *N. farcinica*; *blaAST-1* → amoxicillin-clavulanate resistance in *N. cyriacigeorgica*/*N. neocaledoniensis*; *aph(2'')* → tobramycin resistance; *gyrA* Ser83Ala → ciprofloxacin resistance; 16S rRNA m1A1408 methyltransferase → amikacin resistance ([PMID: 37527397](#)).

Treatment strategy. - **Localized pulmonary/cutaneous:** TMP-SMX (monotherapy can be effective in SOT — favorable outcome in 19/24 patients completing ≥30 days, [PMID: 34143917](#)). - **Severe/disseminated/CNS: combination therapy** (e.g., TMP-SMX + linezolid ± amikacin/carbapenem), often 2–3 drugs initially, then oral step-down; total duration typically **6–12 months** (longer in CNS/immunosuppressed). - **Surgical drainage** of brain and large soft-tissue abscesses improves outcomes. - **Personalized approach:** therapy is **species/susceptibility-guided** via molecular ID and antimicrobial susceptibility testing. - **Pharmacogenomics:** relevant to the sulfonamide component (sulfa hypersensitivity) but no *Nocardia*-specific PGx.

Section 13 — Prevention

- **Primary prevention: TMP-SMX chemoprophylaxis** in high-risk immunosuppressed patients. An individual-patient-data meta-analysis of SOT recipients is titled "*Trimethoprim-sulfamethoxazole significantly reduces the risk of nocardiosis in solid organ transplant recipients*" and assessed "*its dose-response relationship, its effect on preventing disseminated nocardiosis, and the risk of TMP-SMX resistance in case of breakthrough infection*" (PMID: [37865337](#)). Protection is **dose-dependent**: low-dose (thrice-weekly, PJP-dosing) cotrimoxazole "*was not found to prevent nocardiosis*" (PMID: [27090987](#)), whereas higher/daily dosing is protective.
 - **Breakthrough infections** occur despite prophylaxis and in-vitro susceptibility (PMID: [42135975](#)), reflecting host-immunity and drug-exposure factors; prophylaxis **rarely selects resistance** (PMID: [29668123](#)).
 - **Minimize net immunosuppression** where clinically feasible.
 - **Environmental/occupational caution**: reduce soil/dust exposure in severely immunosuppressed patients (gloves, masks for gardening).
 - **De-labeling sulfa allergy** before transplant to preserve access to protective prophylaxis (PMID: [39136148](#)).
 - **Immunization**: No human vaccine exists.
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Section 14 — Other Species / Natural Disease

- **Taxonomy of affected hosts**: Humans (*Homo sapiens*, NCBI:9606) and many animals. *Nocardia* is a natural pathogen of **fish** (*N. seriolae*), **dogs, cattle, and other mammals** (PMID: [4604823](#)).
- **Fish nocardiosis**: *N. seriolae* causes chronic granulomatous disease in >40 species of cultured marine/freshwater fish (largemouth bass, snakehead, amberjack, channel catfish), causing major aquaculture losses (PMID: [38641217](#); PMID: [40827537](#)).
- **Comparative pathology**: Granuloma formation is conserved across teleosts and mammals; teleost granulomas form a macrophage "barrier" via an epithelialization program (PMID: [41443521](#)).
- **Zoonotic potential**: Low/negligible for direct animal-to-human transmission; humans and animals are independently infected from the shared **environmental soil/water reservoir**.

Section 15 — Model Organisms

- **Murine models:** The **C57BL/6 footpad actinomycetoma model** (*N. brasiliensis*) reproduces chronic granulomatous lesions; **eNOS-knockout mice** are protected from actinomycetoma, with increased T-cell proliferation and lower TNF- α — establishing that nitric oxide is required for lesion development (PMID: 33956121).
- **Fish models:** Largemouth bass, snakehead (*Channa argus*), amberjack, and channel catfish infected with *N. seriolae* recapitulate chronic granulomatous disease and macrophage barrier biology; used to study pathogenesis, immune response (IFN- γ), and vaccine/adjuvant development (PMID: 38641217; PMID: 41443521; PMID: 33045332; PMID: 29665356).
- **Phenotype recapitulation:** Both systems reproduce **granuloma formation, macrophage-centered immunity, and chronicity**. Limitations: rodent/fish models do not fully mirror the human iatrogenic-immunosuppression context (transplant, corticosteroids) or human CNS abscess dissemination.

Key Findings (with expanded evidence)

Finding 1 — Opportunistic infection: lung primary, CNS most common secondary site

In a retrospective series of 44 nocardiosis cases, pulmonary infection occurred in 28/44 (64%), disseminated disease in 13/44 (30%), and extrapulmonary single-organ disease in 3/44; 40/44 had comorbidities and 13/44 were on long-term glucocorticoids/immunosuppressants (PMID: 41319540). Predominant isolates were *N. asteroides*, *N. farcinica*, and *N. cyriacigeorgica*. The opportunistic nature is captured succinctly: "*Nocardia are uncommon pathogens that disproportionately afflict the immunocompromised host*" (PMID: 29668123).

Finding 2 — CNS nocardiosis: ~23% case-fatality; *N. farcinica* predominates; surgery improves survival

A systematic review of adult CNS nocardiosis (129 papers) found mean age 55 ± 16 y, 70.8% male, *N. farcinica* the commonest species (39.6%), corticosteroid use the most common predisposing factor (55.8%), brain abscess in 86.9%, and **overall case-fatality 22.8%**; surgery independently improved survival (OR 2.4, 95% CI 0.99–4.11, $p = 0.046$) (PMID: 35700710). Nocardial brain abscesses account for ~2% of all brain abscesses but carry **20–55% mortality** (PMID: 39780099).

Finding 3 — Corticosteroids, transplantation and cell-mediated immune defects dominate risk

In renal transplant recipients, *"High-dose methylprednisolone and cytomegalovirus infection were independent risk factors for Nocardia infection"* (PMID: 39254072). TMP-SMX prophylaxis provides imperfect protection but does not select resistance: *"The use of TMP-SMX prophylaxis was not associated with TMP-SMX-resistant Nocardia"* (PMID: 29668123).

Finding 4 — Nocardiosis affects 0.04–3.5% of SOT recipients; immunosuppression intensity is the driver

The European multicenter case-control study (117 cases/234 controls) identified *"high calcineurin inhibitor trough levels in the month before diagnosis (odds ratio [OR], 6.11; 95% confidence interval [CI], 2.58-14.51), use of tacrolimus (OR, 2.65; 95% CI, 1.17-6.00) and corticosteroid dose (OR, 1.12; 95% CI, 1.03-1.22)"* as independent risk factors, and found *"low-dose cotrimoxazole prophylaxis was not found to prevent nocardiosis"* (PMID: 27090987). One-year mortality was *"10-fold higher in SOT patients with nocardiosis than in those without"* (PMID: 28329348).

Finding 5 — Virulence: ROS-detoxifying enzymes, mycolic-acid wall, proteases, immunosuppressive metabolites

Genomics identified *"espG, mftF, pcaA, fadD32, pks13, narJ, feoB, sodA, katG and mceF"* virulence-associated genes (PMID: 40827537). *N. brasiliensis* *"produces proteases that may play a role in tissue damage, as well as immunosuppressive molecules, such as brasiliardin A"*, and nitric oxide is required for lesions: *"Inflammation and actinomycetoma were prevented in genetically modified [eNOS-knockout] mice infected with N. brasiliensis"* (PMID: 33956121).

Finding 6 — Linezolid universally active; TMP-SMX and amikacin >90%; resistance species-specific

"Linezolid was active against all isolates, followed by trimethoprim/sulfamethoxazole (93%) and amikacin (91%)", with resistance linked to defined determinants such as "blaFAR-1 in N. farcinica (resistance to ceftriaxone)" (PMID: 37527397). An independent Indian cohort confirmed "All tested isolates were susceptible to linezolid and 96 % susceptible to amikacin" (PMID: 40112638).

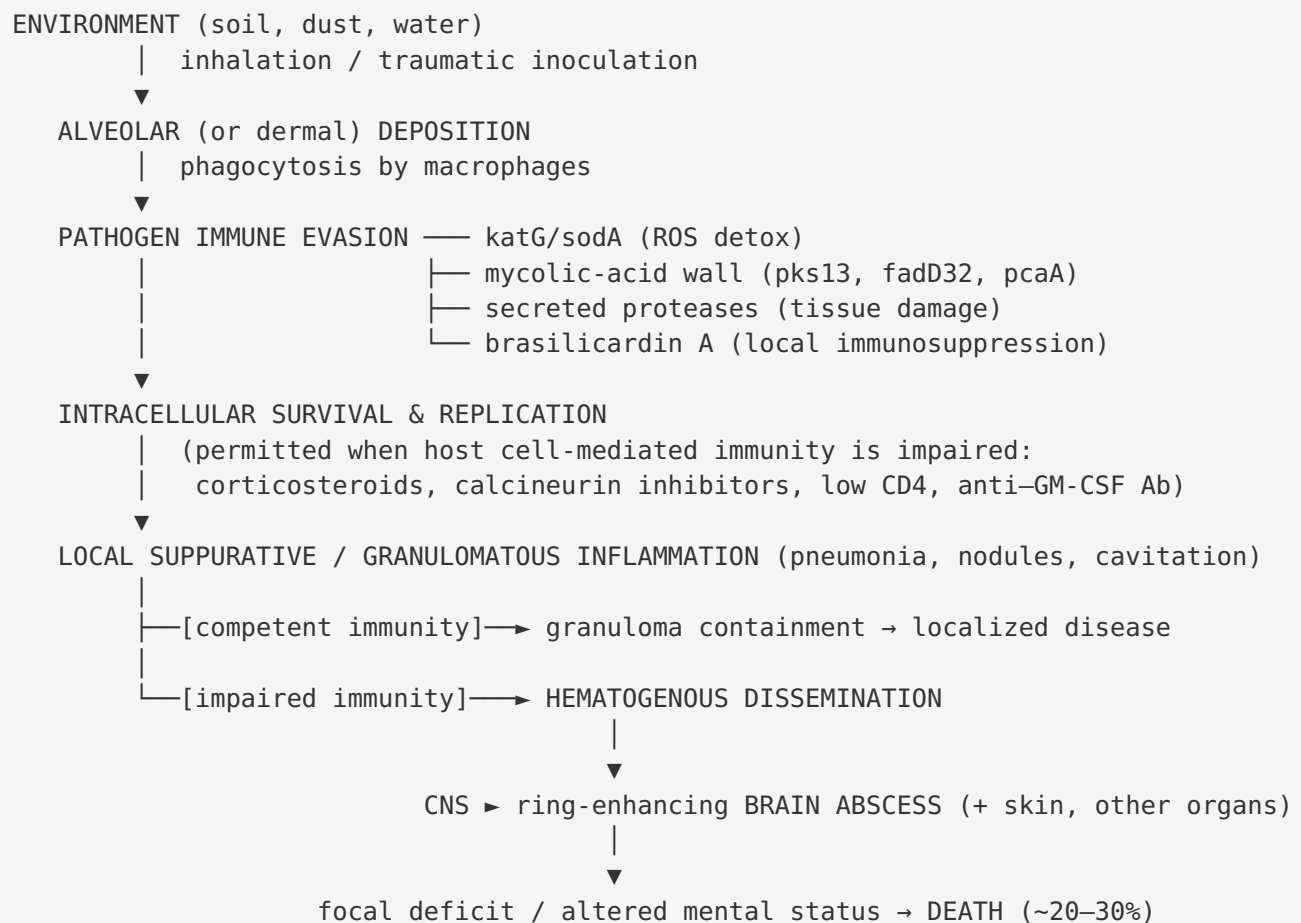
Finding 7 — Diagnosis: microscopy/culture + molecular ID; characteristic CT and MRI findings

Pulmonary CT most commonly shows nodules/masses, ground-glass opacities, consolidation and cavitation — *"Multiple or solitary nodules represented the most common CT feature (n = 32, 94.12%), followed by ground-glass opacities (n = 26, 76.47%), patchy consolidations (n = 25, 73.53%), cavitations (n = 18, 52.94%)"* — with cavitation associated with immunosuppression (*"85% vs 29%, P = 0.005"*) (PMID: 37185005). Molecular species identification uses *"16S rRNA gene sequencing, multilocus sequence analysis (MLSA), matrix-assisted laser desorption ionization-time-of-flight mass spectrometry (MALDI-TOF MS), whole-genome sequencing (WGS), and metagenomic next-generation sequencing (mNGS)"* (PMID: 42258415).

Finding 8 — Prevention: dose-dependent TMP-SMX prophylaxis; rarely selects resistance

The IPD meta-analysis assessed *"its dose-response relationship, its effect on preventing disseminated nocardiosis, and the risk of TMP-SMX resistance in case of breakthrough infection"* (PMID: 37865337), reconciling why *"low-dose cotrimoxazole prophylaxis was not found to prevent nocardiosis"* (PMID: 27090987) — protection is dose-dependent.

Mechanistic Model / Interpretation



The unifying theme is a **two-hit model**: (1) a pathogen equipped to survive inside macrophages and (2) a host whose **cell-mediated immunity is pharmacologically or immunologically disabled**. Upstream events are environmental exposure and immune-evasion virulence; the pivotal branch point is host immune competence, which determines containment versus dissemination. Therapeutic and preventive levers act at defined points: **prophylaxis** (block establishment), **antibiotics** (kill intracellular organisms), **surgery** (evacuate abscess), and **immunosuppression minimization** (restore containment).

Evidence Base

PMID	Study	Contribution
27090987	European SOT case-control (117/234)	Quantifies immunosuppression risk factors; low-dose prophylaxis inadequate
28329348	European SOT outcomes	10-fold excess 1-year mortality
29668123	Transplant review (Duke cohort)	Opportunistic nature; prophylaxis does not select resistance
35700710	CNS systematic review (129 papers)	22.8% case-fatality; <i>N. farcinica</i> ; surgery benefit
37527397	WGS of 138 isolates (Israel)	Linezolid 100%; gene-encoded resistance
40112638	Indian susceptibility series	Independent confirmation of linezolid/amikacin activity
37185005	CT of pulmonary nocardiosis	CT feature frequencies; cavitation ↔ immunosuppression
42258415	ID/AST review	Molecular diagnostic methods
37865337	IPD meta-analysis	Dose-dependent TMP-SMX prevention
40827537	<i>N. seriolae</i> genomics	Virulence genes (<i>sodA</i> , <i>katG</i> , mycolic-acid)
33956121	eNOS-KO mouse model	Proteases, <i>brasilicardin A</i> ; NO required for lesions
39254072	Renal transplant (Pakistan)	Methylprednisolone, CMV as independent risks
39780099	CNS abscess case/review	Brain-abscess frequency and 20–55% mortality
36303280	SOT cohort (n=125)	12-mo mortality; liver Tx and delay as prognostic factors
34143917	TMP-SMX monotherapy in SOT	Monotherapy effective in most SOT cases
39136148	Sulfa allergy label cohort	Sulfa label ↑ nocardiosis risk (HR 3.85)
22825801	<i>N. brasiliensis</i> immunology	Systemic immunity vs local immunosuppressive microenvironment
41443521	Teleost granuloma study	Conserved macrophage-barrier granuloma biology

Limitations and Knowledge Gaps

1. **Template mismatch.** This template assumes a heritable/Mendelian disease. For an infectious disease, sections on causal human genes, inheritance, penetrance, germline variant classification, and genetic screening are **Not Applicable**; molecular content necessarily shifts to pathogen biology.
 2. **Epidemiology is under-characterized.** True population incidence/prevalence is uncertain because nocardiosis is not reportable in most jurisdictions and is under-diagnosed. Regional species-distribution and burden data are inadequate in many settings.
 3. **Evidence quality.** Much of the clinical literature is retrospective case series and single-center cohorts; randomized treatment trials are essentially absent. Treatment durations and regimens are guided by expert consensus, not RCTs.
 4. **Host-side mechanism is inferred.** The requirement for intact cell-mediated immunity is established by clinical association and animal models; direct human mechanistic proof (which T-cell/macrophage pathways fail under specific drugs) is incomplete.
 5. **In-vitro/in-vivo discordance.** Breakthrough disease despite in-vitro susceptibility ([PMID: 42135975](#)) shows susceptibility testing imperfectly predicts outcome.
 6. **Species-level resistance heterogeneity.** *N. otitidiscaviarum* and inter-strain variability complicate empiric therapy ([PMID: 42143233](#)).
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Proposed Follow-up Experiments / Actions

1. **Prospective incidence registry** stratified by immunosuppression regimen to define modern population burden and optimal prophylaxis dosing/duration.
2. **Randomized/adaptive trial of prophylaxis dosing** (single-strength daily vs thrice-weekly) in SOT/HCT to formalize the dose-response signal from [PMID: 37865337](#).
3. **WGS-linked antimicrobial-resistance surveillance** to build species→resistance prediction rules (extending *blaFAR-1/blaAST-1/gyrA/16S-methyltransferase* mapping) enabling genotype-guided empiric therapy before phenotypic AST returns.
4. **Comparative-effectiveness study of TMP-SMX monotherapy vs combination** for disseminated/CNS disease, powered for mortality (current analyses are underpowered, [PMID: 36303280](#)).

5. **Host-immunity biomarkers** (CD4 recovery, anti-GM-CSF autoantibody screening) to risk-stratify and target prophylaxis.
 6. **Rapid molecular diagnostics** (mNGS/tNGS) validation to shorten time-to-species-ID and drainage decisions.
 7. **Mechanistic dissection** of macrophage killing defects under calcineurin inhibitors/corticosteroids using human macrophage-*Nocardia* infection models to connect the two-hit model to actionable host-directed therapy.
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Report compiled from 8 confirmed findings and 59 reviewed papers across 5 investigation iterations. Evidence types span human clinical cohorts/case-control studies, systematic reviews, pathogen genomics, and murine/teleost model-organism studies.

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